

Alexey Vavilin

Senior Software Engineer

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SUMMARY

Backend and systems developer with 15 years of experience, focused on Rust for the last 7+ years. I've worked on blockchain node components (TON/Everscale), SDKs for client-node interaction, data processing pipelines for Filecoin, and — earlier in my career — Windows kernel-mode drivers and cryptographic software. Comfortable across the stack: performance tuning, low-level systems work, applied cryptography, and general backend architecture — including service orchestration and task-pipeline design. I have deep expertise in low-level code optimization — profiling and reworking systems for throughput, latency, and resource usage. I actively use AI tools in my day-to-day work, and I enjoy learning new technologies and exploring new domains.

EXPERIENCE

RSquad

Senior Rust Developer | June 2025 – Present

Building and optimizing components of a Rust-based TON blockchain node as part of a dedicated team. Work spans both new feature development and performance optimization of existing subsystems (memory usage, disk I/O, transaction throughput).

- Implemented an archive node mode storing the network's full block history (20 TB) and historical state (8 TB).
- Optimized account data statistics calculation from 500 ms to 50 ms for the largest possible account size.

web3mine

Software Engineer | November 2023 – May 2025

- Implemented an efficient alternative solution in Rust for data onboarding to the Filecoin network, optimized for CPU, disk, and memory usage, and leveraging the latest blockchain capabilities to reduce gas consumption. Improved data processing throughput from 3 GB/s (limited by the Linux file cache) to 20 GB/s using Direct IO and optimized task parallelization.
- Designed and built a task dispatcher orchestrating a pipeline of onboarding services, each handling its own processing stage, supporting up to 50,000 active tasks running concurrently.

EverX

Rust Developer | November 2018 – October 2023

- Built the SDK for TON/Everscale blockchain clients: node interoperation, transaction submission, result querying, local VM emulation, ABI encoding/decoding, encryption and hashing utilities. Supported Linux, macOS, Windows, and WASM targets, with third-party language bindings maintained for 11 languages.
- Developed blockchain indexing services processing live data streams for explorers, token wallets, and decentralized exchanges, handling up to 100,000 TPS in performance testing.
- Contributed to TON and Everscale blockchain node components: transaction executor, block producer, chain state manager.

Cryptosoft

System Developer | November 2008 – October 2018

Developed cryptographic data-protection software for Windows: OS kernel-mode drivers and client API DLLs for driver interoperation. Performed analysis and reverse engineering of system executables.

SKILLS

Languages: Rust, Go, Python, C/C++

Blockchain & Protocols: Blockchain, TON / TVM, Filecoin / IPFS

Data & Infrastructure: PostgreSQL, RocksDB, Kafka, WASM, Docker, Kubernetes

Systems & Security: Cryptography, System programming, Windows kernel drivers

Engineering Practices: Performance optimization, Distributed systems

AI Tools: Claude Code, Codex, Cursor

EDUCATION

Penza State University — Engineer's degree, Computer Software Engineering

2005 – 2010